

Experiment - 11

Query with Joins – EquiJoin, InnerJoin, OuterJoin

```
Database changed
mysql> CREATE TABLE Department
-> (
-> DeptID INT PRIMARY KEY,
-> DeptName VARCHAR(50)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Department VALUES
-> (1, 'CSE'),
-> (2, 'ECE'),
-> (3, 'IT'),
-> (4, 'Mechanical');
Query OK, 4 rows affected (0.02 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Department;
```

DeptID	DeptName
1	CSE
2	ECE
3	IT
4	Mechanical

```
4 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Faculty
-> (
-> FacultyID INT PRIMARY KEY,
-> FacultyName VARCHAR(50),
-> DeptID INT,
-> FOREIGN KEY(DeptID) REFERENCES Department(DeptID)
-> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> INSERT INTO Faculty VALUES
-> (101,'Kamal',2),
-> (102,'Rahul',1),
-> (103,'Anitha',3),
-> (104,'Ravi',2);
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Faculty;
+-----+-----+-----+
| FacultyID | FacultyName | DeptID |
+-----+-----+-----+
| 101 | Kamal | 2 |
| 102 | Rahul | 1 |
| 103 | Anitha | 3 |
| 104 | Ravi | 2 |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Student
-> (
-> StudentID INT PRIMARY KEY,
-> StudentName VARCHAR(50)
-> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> INSERT INTO Student VALUES
-> (1,'Indhu'),
-> (2,'Rahul'),
-> (3,'Priya'),
-> (4,'Sneha');
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Student;
+-----+-----+
| StudentID | StudentName |
+-----+-----+
| 1 | Indhu |
| 2 | Rahul |
| 3 | Priya |
| 4 | Sneha |
+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Course
-> (
-> CourseNo VARCHAR(10) PRIMARY KEY,
-> CourseName VARCHAR(50),
-> FacultyID INT,
-> FOREIGN KEY(FacultyID) REFERENCES Faculty(FacultyID)
-> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> INSERT INTO Course VALUES
-> ('CS101','DBMS',102),
-> ('EC201','Digital Electronics',101),
-> ('EC202','Signals',104),
-> ('IT301','Cloud Computing',103);
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Course;
+-----+-----+-----+
| CourseNo | CourseName | FacultyID |
+-----+-----+-----+
| CS101 | DBMS | 102 |
| EC201 | Digital Electronics | 101 |
| EC202 | Signals | 104 |
| IT301 | Cloud Computing | 103 |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Score
-> (
-> StudentID INT,
-> CourseNo VARCHAR(10),
-> Marks INT,
-> FOREIGN KEY(StudentID) REFERENCES Student(StudentID),
-> FOREIGN KEY(CourseNo) REFERENCES Course(CourseNo)
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> INSERT INTO Score VALUES
-> (1,'CS101',90),
-> (2,'EC201',85),
-> (3,'IT301',95);
```

Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM Score;
```

StudentID	CourseNo	Marks
1	CS101	90
2	EC201	85
3	IT301	95

3 rows in set (0.00 sec)

```
mysql> SELECT Student.StudentName,
-> Course.CourseName
-> FROM Student
-> CROSS JOIN Course;
```

StudentName	CourseName
Sneha	DBMS
Priya	DBMS
Rahul	DBMS
Indhu	DBMS
Sneha	Digital Electronics
Priya	Digital Electronics
Rahul	Digital Electronics
Indhu	Digital Electronics
Sneha	Signals
Priya	Signals
Rahul	Signals
Indhu	Signals
Sneha	Cloud Computing
Priya	Cloud Computing
Rahul	Cloud Computing
Indhu	Cloud Computing

16 rows in set (0.01 sec)

```
mysql> SELECT Faculty.FacultyName,
-> Department.DeptName
-> FROM Faculty, Department
-> WHERE Faculty.DeptID = Department.DeptID;
```

FacultyName	DeptName
Kamal	ECE
Rahul	CSE
Anitha	IT
Ravi	ECE

4 rows in set (0.00 sec)

```
mysql> SELECT Faculty.FacultyName,
-> Department.DeptName
-> FROM Faculty
-> INNER JOIN Department
-> ON Faculty.DeptID = Department.DeptID;
```

FacultyName	DeptName
Kamal	ECE
Rahul	CSE
Anitha	IT
Ravi	ECE

4 rows in set (0.00 sec)

```
mysql> SELECT Student.StudentName,
-> Score.CourseNo,
-> Score.Marks
-> FROM Student
-> LEFT JOIN Score
-> ON Student.StudentID = Score.StudentID;
```

StudentName	CourseNo	Marks
Indhu	CS101	90
Rahul	EC201	85
Priya	IT301	95
Sneha	NULL	NULL

4 rows in set (0.00 sec)

```
mysql> SELECT Faculty.FacultyName,
-> Department.DeptName
-> FROM Faculty
-> INNER JOIN Department
-> ON Faculty.DeptID = Department.DeptID;
```

FacultyName	DeptName
Kamal	ECE
Rahul	CSE
Anitha	IT
Ravi	ECE

4 rows in set (0.00 sec)

```
mysql> SELECT Student.StudentName,
-> Score.CourseNo
-> FROM Student
-> LEFT JOIN Score
-> ON Student.StudentID = Score.StudentID
-> WHERE Score.CourseNo IS NULL;
```

StudentName	CourseNo
Sneha	NULL

1 row in set (0.00 sec)

```
mysql> SELECT Course.CourseName
-> FROM Course
-> INNER JOIN Faculty
-> ON Course.FacultyID = Faculty.FacultyID
-> INNER JOIN Department
-> ON Faculty.DeptID = Department.DeptID
-> WHERE Faculty.FacultyName='Kamal'
-> AND Department.DeptName='ECE';
```

CourseName
Digital Electronics

1 row in set (0.00 sec)

```
mysql> SELECT * FROM Department;
```

DeptID	DeptName
1	CSE
2	ECE
3	IT
4	Mechanical

```
4 rows in set (0.00 sec)
```

```
mysql>
mysql> SELECT * FROM Faculty;
```

FacultyID	FacultyName	DeptID
101	Kamal	2
102	Rahul	1
103	Anitha	3
104	Ravi	2

```
4 rows in set (0.00 sec)
```

```
mysql>
mysql> SELECT * FROM Student;
```

StudentID	StudentName
1	Indhu
2	Rahul
3	Priya
4	Sneha

```
4 rows in set (0.00 sec)
```

```
mysql>
mysql> SELECT * FROM Course;
```

CourseNo	CourseName	FacultyID
CS101	DBMS	102
EC201	Digital Electronics	101
EC202	Signals	104
IT301	Cloud Computing	103

```
4 rows in set (0.00 sec)
```

```
mysql>
mysql> SELECT * FROM Score;
```

StudentID	CourseNo	Marks
1	CS101	90
2	EC201	85
3	IT301	95

```
3 rows in set (0.00 sec)
```